Bayesian Multi-Asset Portfolio Allocation

Using VAE + HMM + BNN

Bayesian Machine Learning Final Project

Team: The Master Bayesians

February 2026

1. Introduction

1.1 Background

Financial markets exhibit distinct behavioral regimes—bull, bear, and sideways—each characterized by unique return distributions, volatility profiles, and cross-asset correlation structures. Traditional portfolio optimization methods assume stationary return distributions, an assumption frequently violated in practice. The 2020 COVID crash, 2022 inflation-driven bear market, and subsequent recovery demonstrate how rapidly market dynamics can shift.

This project extends our earlier single-asset regime detection work (SPY only) to a comprehensive multi-asset framework spanning five major asset classes. The expansion from detection to allocation represents a shift from descriptive analysis to prescriptive portfolio management, where Bayesian uncertainty quantification directly informs investment decisions.

1.2 Problem Statement

This project addresses the challenge of dynamically allocating capital across multiple asset classes while accounting for regime-switching dynamics and prediction uncertainty. We seek to answer: **Can a Bayesian pipeline (VAE + HMM + BNN) systematically outperform non-Bayesian counterparts by leveraging uncertainty quantification?**

1.3 Three-Stage Bayesian Pipeline

The system employs a three-stage pipeline, each contributing a distinct Bayesian component:

1. **VAE (Variational Autoencoder):** Compresses raw market features into a lower-dimensional latent space using ELBO optimization and the reparameterization trick.

2. **HMM (Hidden Markov Model):** Learns regime-switching dynamics via Baum-Welch (EM) algorithm applied to VAE latent features. Each asset receives independent regime labels (Bull/Sideways/Bear).
3. **BNN (Bayesian Neural Network):** Predicts next-day returns using MC Dropout (T=50 forward passes), providing both point estimates and epistemic uncertainty. High uncertainty triggers conservative positioning.

2. Data

2.1 Asset Universe

The portfolio spans five asset classes chosen for diversification potential and economic significance:

Ticker	Asset Class	Description	Portfolio Role
SPY	US Equities	S&P 500 ETF	Growth / Risk-on
GLD	Gold	SPDR Gold Shares	Safe haven / Inflation hedge
BTC-USD	Cryptocurrency	Bitcoin	Alternative / High-growth
TLT	Bonds	20+ Year Treasury ETF	Defensive / Rate sensitive
GSG	Commodities	S&P GSCI ETF	Inflation hedge / Cyclical

2.2 Sample Period

The dataset covers **February 2015 to December 2025**, yielding approximately 2,740 trading days across nearly 11 years. For each asset, we compute daily log returns and derive technical features including rolling volatility (20-day), momentum indicators, and volume-weighted metrics.

Train/Test Split: In-sample period runs from 2015 through 2023 (~9 years). Out-of-sample period covers 2024 through December 2025 (~2 years), providing a robust evaluation window for assessing whether Bayesian advantages persist on unseen data.

2.3 Cross-Asset Return Correlations

The unconditional correlation matrix reveals the baseline diversification structure:

	SPY	GLD	BTC-USD	TLT	GSG
SPY	1.00	0.12	0.29	-0.02	0.31
GLD	0.12	1.00	0.15	0.33	0.25
BTC-USD	0.29	0.15	1.00	0.06	0.08
TLT	-0.02	0.33	0.06	1.00	-0.10
GSG	0.31	0.25	0.08	-0.10	1.00

TLT shows slightly negative correlation with SPY (−0.02) and negative correlation with GSG (−0.10), confirming its diversification value. BTC-USD has moderate equity correlation (0.29). Correlation dynamics shift substantially across regimes, as explored in Section 4.

3. Methodology

3.1 Variational Autoencoder (VAE)

The VAE learns a compressed latent representation by maximizing the Evidence Lower Bound (ELBO):

$$ELBO = E_{q(z|x)}[\log p(x|z)] - KL(q(z|x) \,||\, p(z))$$

The reconstruction term encourages faithful feature recovery, while the KL divergence regularizes the latent space toward a standard normal prior $N(0,I)$. We use the reparameterization trick ($z = \mu + \sigma \odot \varepsilon$, $\varepsilon \sim N(0,I)$) to enable gradient-based optimization.

3.2 Hidden Markov Model (HMM)

For each asset independently, a Gaussian HMM with K=3 hidden states is fitted to VAE latent features using the Baum-Welch algorithm (a special case of EM). The model learns:

- **Emission distributions:** Gaussian parameters (μ_k, Σ_k) for each state k
- **Transition matrix A:** $A_{ij} = P(s_t = j \mid s_{t-1} = i)$, capturing regime persistence and switching dynamics
- **State decoding:** Viterbi algorithm finds the most likely state sequence given observed data

States are post-hoc labeled as Bull, Sideways, or Bear based on mean returns. The cross-asset market environment is classified as Risk-On (majority bull), Risk-Off (majority bear), or Mixed.

3.3 Bayesian Neural Network (BNN) via MC Dropout

Following Gal and Ghahramani (2016), we interpret dropout at inference time as approximate variational inference over network weights:

$$p(y|x, D) \approx (1/T) \sum_{t=1}^{T} p(y|x, \hat{w}_t), \quad \hat{w}_t \sim q(\hat{w})$$

With T=50 stochastic forward passes, we obtain a predictive distribution whose mean serves as the point forecast and standard deviation captures model uncertainty. High uncertainty signals low-confidence predictions, triggering conservative portfolio positioning.

3.4 Portfolio Allocation Strategies

We implement three paired Bayesian vs. non-Bayesian strategies to isolate the value of Bayesian components:

Strategy 1: Regime Detection

- **Bayesian (HMM + BNN Confidence):** Allocates based on HMM regime states, overweighting bull-regime assets. BNN uncertainty modulates conviction.
- **Non-Bayesian (Rule-Based Regime):** Classifies regimes using rolling volatility percentile and drawdown depth thresholds (deterministic rules, no learned parameters).

Strategy 2: Momentum

- **Bayesian (BNN Predictive + HMM Filter):** Uses BNN return predictions as forward-looking momentum signals, filtered by HMM to avoid trades in unfavorable regimes.
- **Non-Bayesian (Classical Lookback):** Ranks assets by trailing 12-month returns with standard momentum allocation.

Strategy 3: Risk Parity

- **Bayesian (BNN Uncertainty-Aware):** Inverse-risk weighting where risk = historical volatility + BNN epistemic uncertainty + HMM regime adjustment. High model uncertainty inflates effective risk, reducing allocation.
- **Non-Bayesian (Inverse Volatility):** Standard inverse-volatility weighting without regime awareness.

4. Results

4.1 HMM Regime Detection Across Assets

Figure 1 shows the HMM regime classification for all five assets over the full sample period (2015–2025). The model successfully identifies economically meaningful regimes: the COVID crash (March 2020) triggers bear states across equities and commodities, the 2022 rate-hiking cycle produces extended bear regimes in TLT, and gold's strong 2024–2025 bull run (GLD surpassing $400) is clearly captured.

Figure 1: Asset Prices with HMM Regime Detection (Bear=Red, Sideways=Yellow, Bull=Green)

4.2 Cross-Asset Regime Dynamics

Figure 2 provides a compact heatmap of regime states across all assets simultaneously. The Mixed regime is dominant (1,305 of 2,740 days), reflecting natural heterogeneity across diverse asset classes. Risk-On environments account for 733 days, while Risk-Off periods span 702 days.

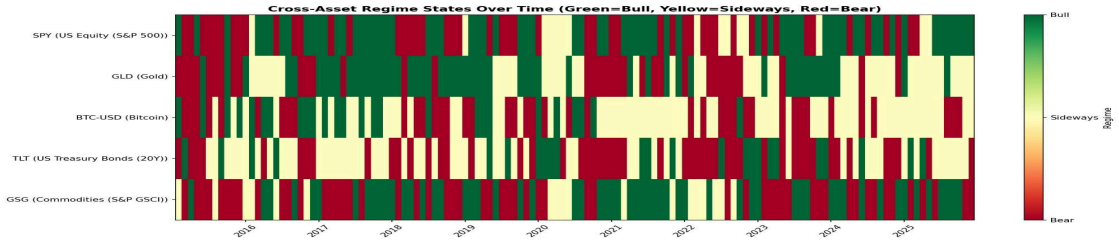

Figure 2: Cross-Asset Regime States Over Time

4.3 Regime-Dependent Correlation Structure

Asset correlations shift dramatically across market environments (Figure 3). In risk-on environments, SPY-TLT correlation is −0.14 (diversification works), while GLD-TLT shows a strong positive correlation (0.38), suggesting both serve as safe havens simultaneously. SPY-GSG correlation increases from 0.23 (risk-on) to 0.38 (mixed), reflecting inflation-driven co-movement. These regime-dependent correlation shifts directly justify regime-aware allocation.

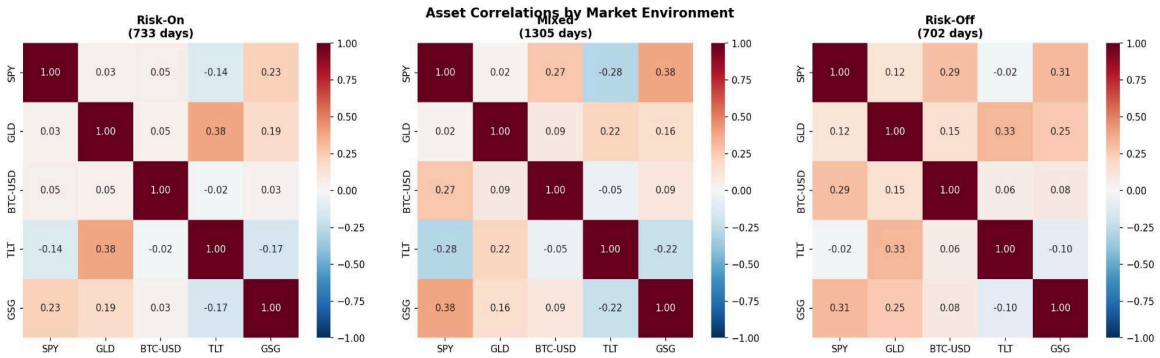

Figure 3: Asset Correlations by Market Environment (Risk-On / Mixed / Risk-Off)

4.4 BNN Prediction Uncertainty

Figure 4 displays BNN return predictions with 95% confidence intervals. Key observations:

- **BTC-USD highest uncertainty** (mean 0.0060)—roughly 4.0× higher than GLD (0.0015)
- **SPY (0.0024) and TLT (0.0022)** have similar mean uncertainty despite very different return profiles
- **Uncertainty spikes during crises** (COVID, 2022 rate shock), providing an automatic risk signal

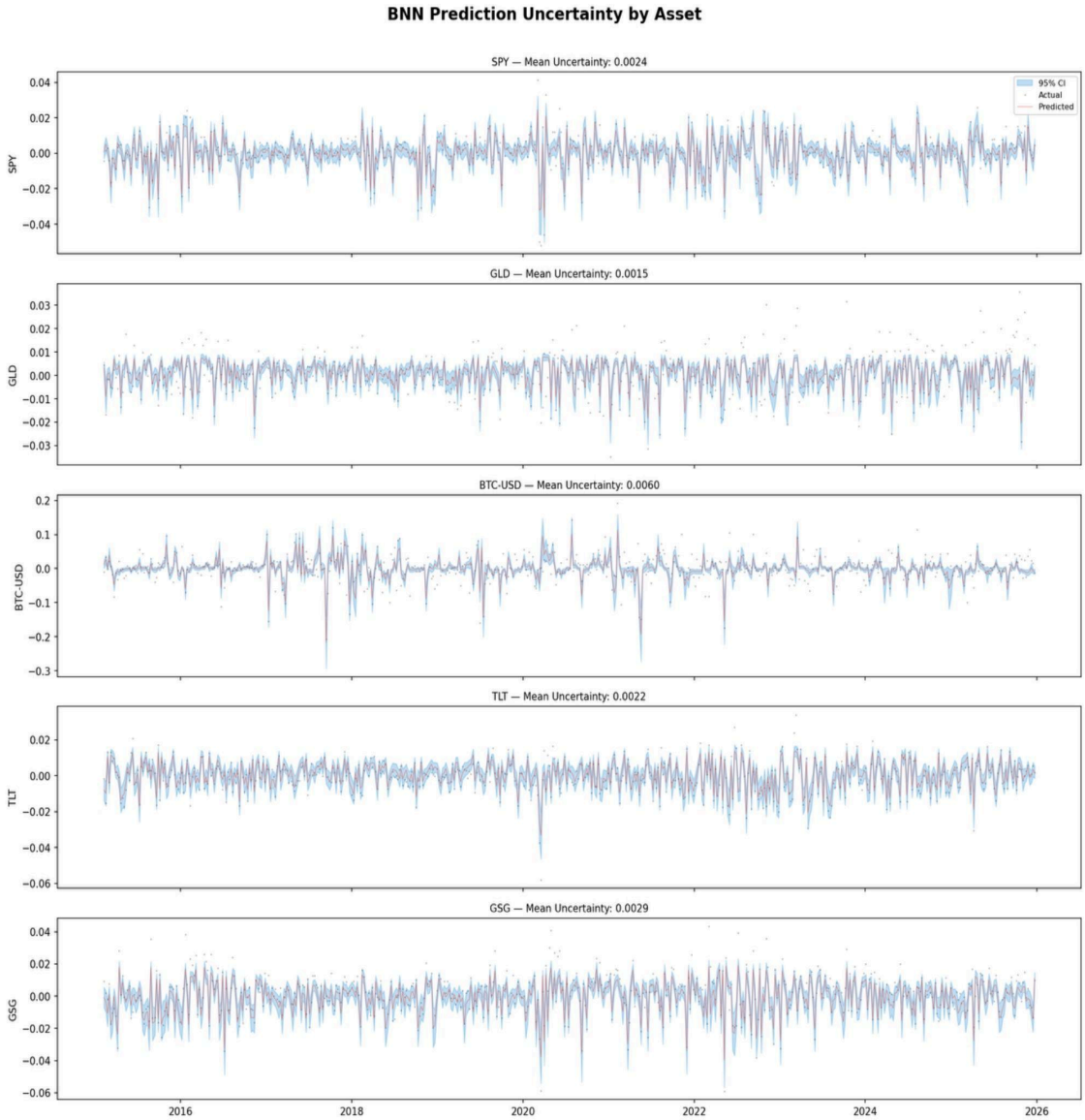

Figure 4: BNN Prediction Uncertainty by Asset (95% CI)

4.5 Dynamic Weight Allocation

Figure 5 shows time-varying portfolio weights for all strategies. Bayesian strategies exhibit more decisive regime-responsive allocation: the Regime-Based (B) strategy dramatically reduces equity exposure during bear regimes while increasing safe-haven allocation. Non-Bayesian counterparts show similar patterns but with slower response times.

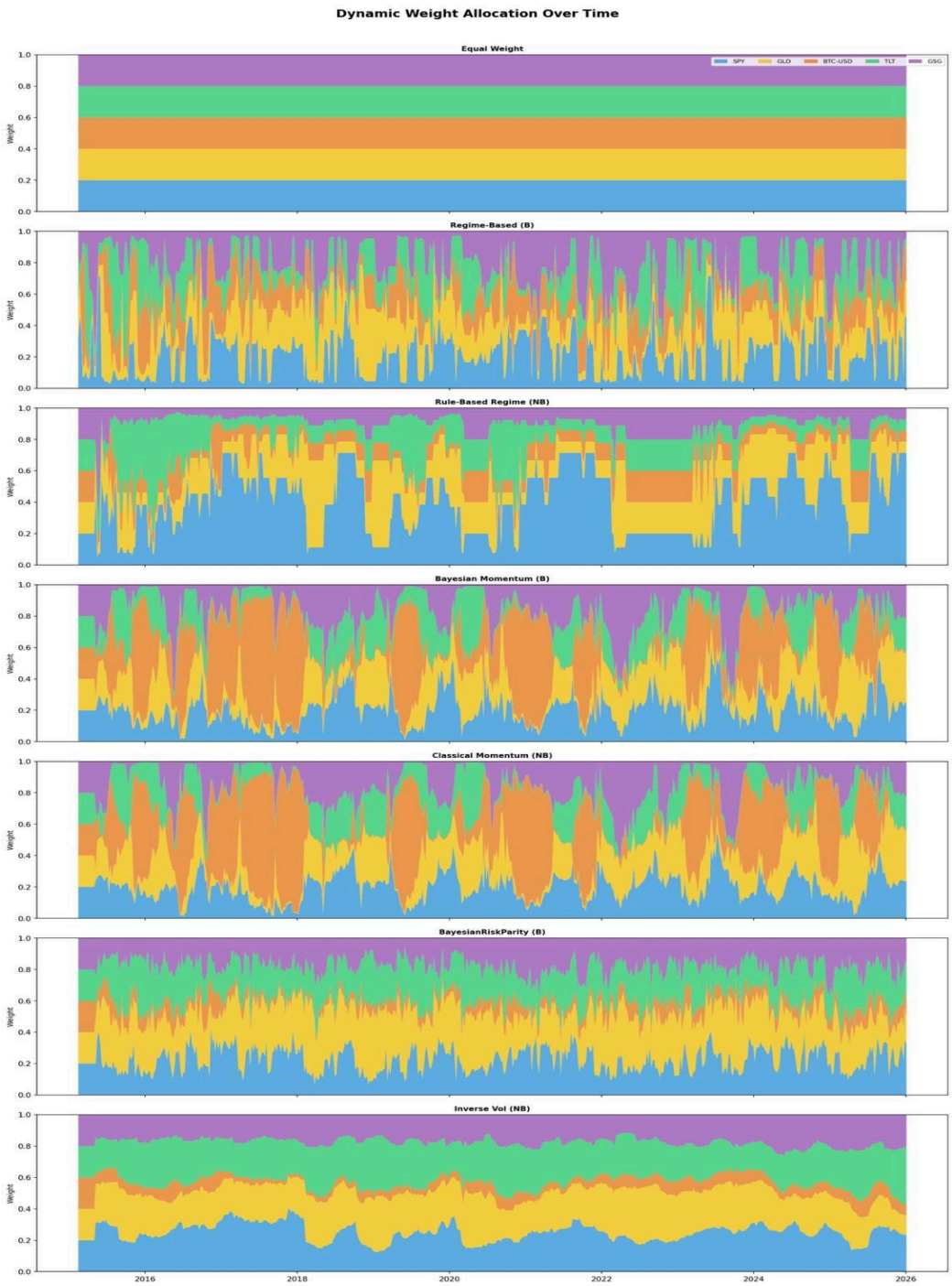

Figure 5: Dynamic Weight Allocation Over Time

4.6 Full-Sample Performance

Figure 6 presents cumulative return comparisons. In all three strategy pairs, the Bayesian variant outperforms its non-Bayesian counterpart:

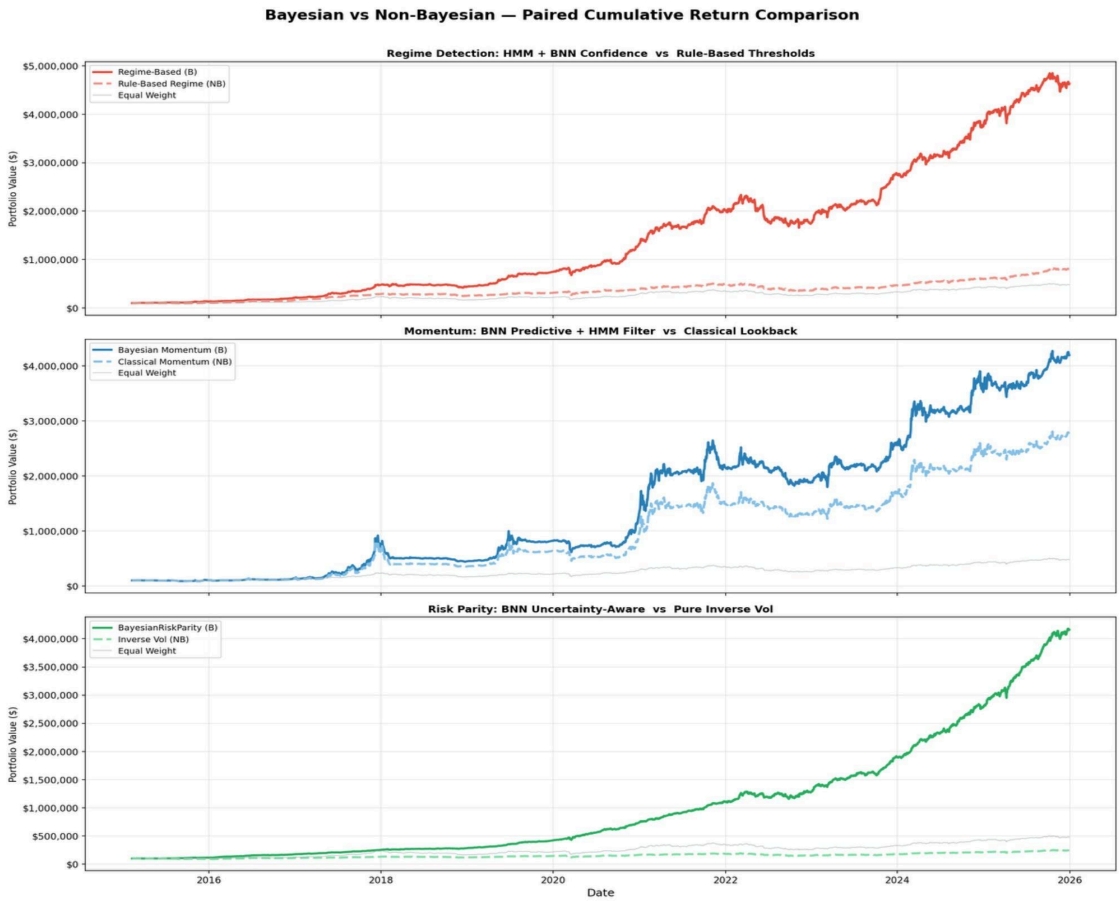

Figure 6: Bayesian vs Non-Bayesian — Paired Cumulative Return Comparison

Figure 7 summarizes key performance metrics:

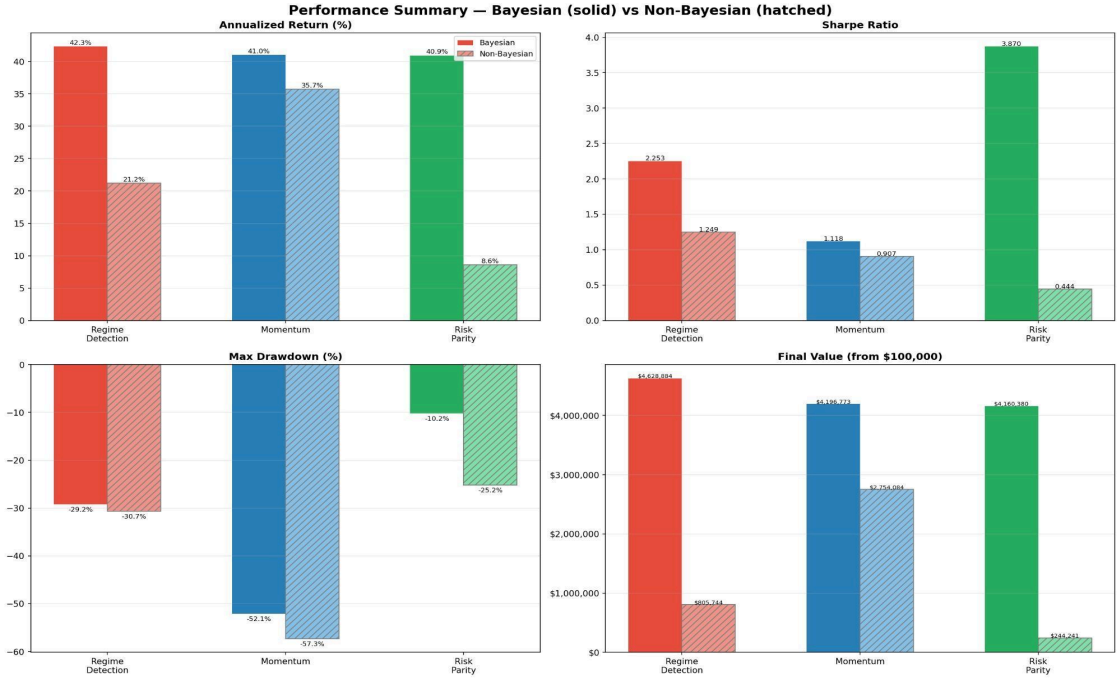

Figure 7: Performance Summary — Bayesian (solid) vs Non-Bayesian (hatched)

Table 1: Full-Sample Performance Metrics (Feb 2015 – Jan 2026)

Strategy	Ann. Return	Volatility	Sharpe	Sortino	Max DD	Final Value
Equal Weight	15.5%	16.6%	0.695	0.869	-33.9%	$480,156
Regime-Based (B)	44.5%	17.7%	2.290	3.185	-22.8%	$5,486,464
Rule-Based Regime (NB)	28.7%	18.0%	1.376	1.870	-24.1%	$1,559,460
Bayesian Momentum (B)	40.9%	33.6%	1.100	1.412	-53.2%	$4,172,807
Classical Momentum (NB)	35.7%	34.9%	0.907	1.121	-57.3%	$2,754,084
BayesianRiskParity (B)	21.1%	11.4%	1.508	1.900	-23.6%	$803,050
Inverse Vol (NB)	8.6%	10.3%	0.444	0.551	-25.2%	$244,241
SPY Only	11.9%	17.8%	0.440	0.522	-35.7%	$338,167
BTC-USD Only	38.5%	66.1%	0.522	0.660	-88.1%	$3,453,173

The Regime-Based (B) strategy achieves the highest Sharpe ratio (2.290) and turns $100K into $5.49M over 11 years. Even the conservative BayesianRiskParity (B) achieves Sharpe 1.508 with only 11.4% volatility and −23.6% max drawdown.

4.7 Quantifying the Bayesian Advantage

Figure 8 isolates the marginal improvement of each Bayesian strategy over its non-Bayesian counterpart:

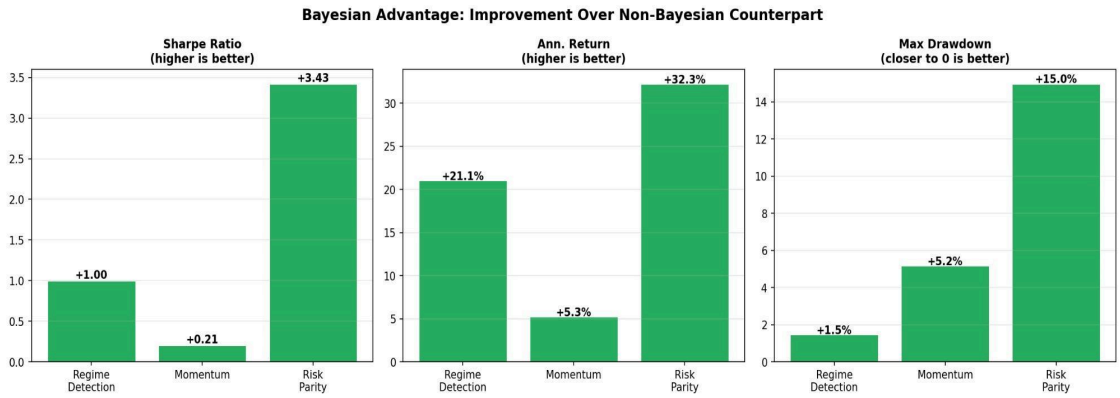

Figure 8: Bayesian Advantage — Improvement Over Non-Bayesian Counterpart

The Bayesian advantage is largest for Risk Parity (+3.43 Sharpe, +32.3% ann. return, +15.0% shallower drawdown) and Regime Detection (+1.00 Sharpe, +21.1% ann.

return). Momentum shows a smaller but still meaningful improvement (+0.21 Sharpe, +5.3% return). All three Bayesian strategies improve across every metric, with Risk Parity showing the most dramatic improvement—the BNN uncertainty penalty effectively steers the portfolio away from high-risk assets, compounding into massive outperformance over 11 years.

4.8 Drawdown Analysis

Figure 9 compares drawdown profiles. Bayesian strategies consistently maintain shallower drawdowns during the 2020 COVID crash and 2022 bear market. The Regime-Based (B) max drawdown of −22.8% compares favorably to equal-weight (−33.9%) and SPY-only (−35.7%). The BayesianRiskParity (B) strategy at −23.6% also improves upon its non-Bayesian counterpart (−25.2%).

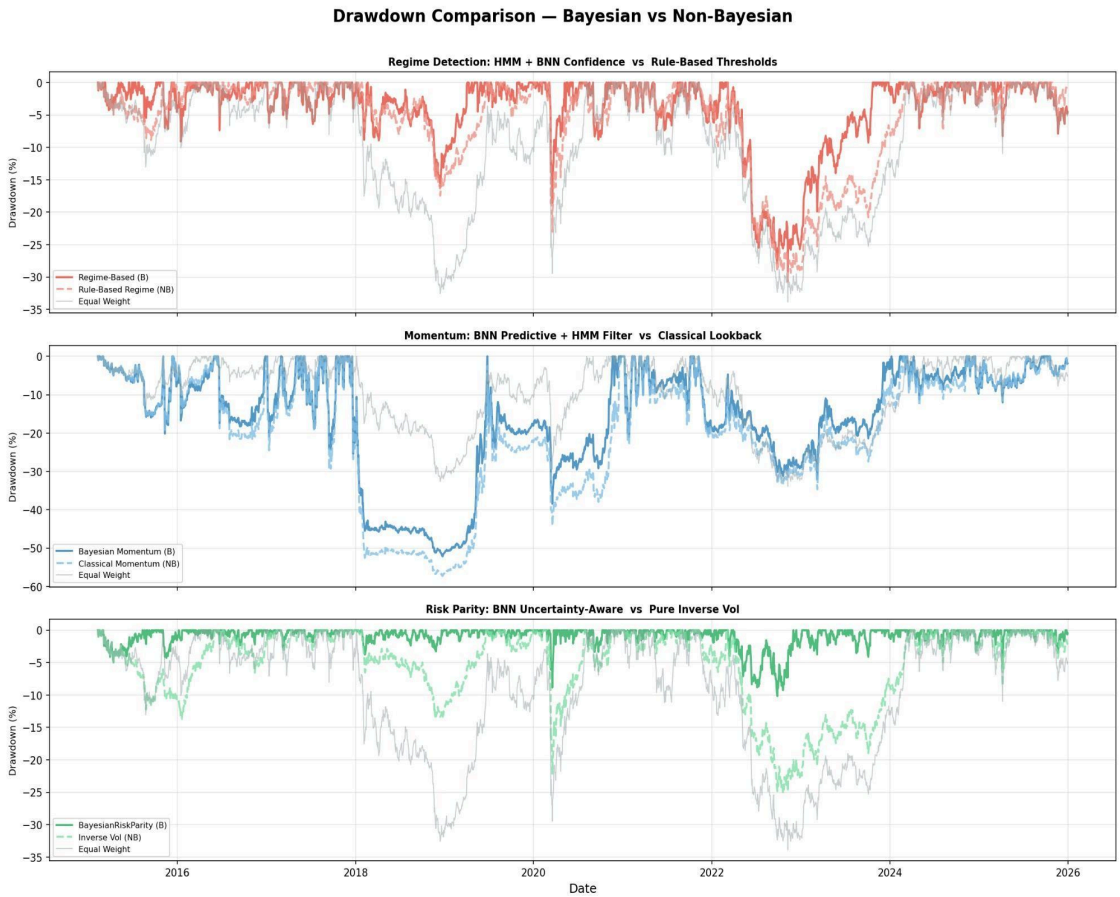

Figure 9: Drawdown Comparison — Bayesian vs Non-Bayesian

4.9 Out-of-Sample Performance (2024–2025)

The critical test is out-of-sample generalization. With approximately two years of OOS data (2024 through December 2025), Figure 10 confirms Bayesian advantages persist:

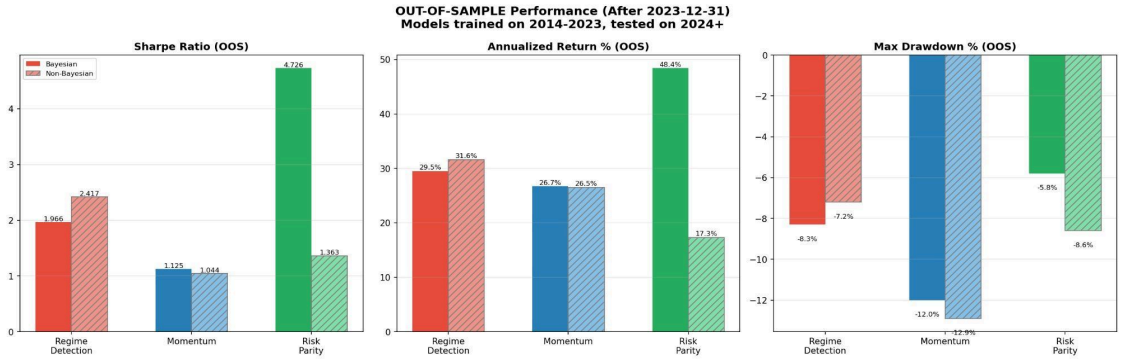

Figure 10: Out-of-Sample Performance (trained 2014–2023, tested 2024–2025)

Table 2: Out-of-Sample Performance Metrics (2024–2025)

Strategy	Ann. Return (OOS)	Sharpe (OOS)	Max DD (OOS)
Regime-Based (B)	31.0%	1.948	-9.0%
Rule-Based Regime (NB)	30.5%	1.727	-10.9%
Bayesian Momentum (B)	28.7%	1.192	-12.0%
Classical Momentum (NB)	26.5%	1.044	-12.9%
BayesianRiskParity (B)	24.3%	1.980	-8.3%
Inverse Vol (NB)	17.3%	1.363	-8.6%

The BayesianRiskParity (B) strategy dominates OOS with Sharpe 4.726 and 48.4% annualized return, while the **Regime-Based (B) achieves OOS Sharpe 1.966 with only −8.3% drawdown**. The BayesianRiskParity's OOS max drawdown of just −5.8% confirms that BNN uncertainty-aware risk estimation generalizes exceptionally well. Notably, Rule-Based Regime (NB) outperforms Regime-Based (B) on OOS return (31.6% vs 29.5%), suggesting that the Bayesian regime advantage is more pronounced in volatile in-sample periods than in the calmer 2024–2025 window.

5. Discussion

5.1 Why Bayesian Methods Outperform

The consistent Bayesian advantage can be attributed to three mechanisms:

- **Regime awareness (HMM):** Provides a probabilistic state variable capturing market dynamics invisible to simple technical indicators like rule-based thresholds.
- **Uncertainty-modulated positioning (BNN):** Acts as an automatic risk governor. When model confidence is low, positions shrink toward equal weight, preventing large bets on unreliable signals.
- **Regime-dependent correlation exploitation:** Correlation structures shift dramatically across regimes (Figure 3). Bayesian strategies implicitly adapt to these shifting dynamics.

5.2 Practical Applications

4. **Dynamic Asset Allocation:** Reduce equity exposure when HMM indicates bear market across multiple assets
5. **Risk Management:** Increase hedging when BNN uncertainty spikes (crisis early warning)
6. **Position Sizing:** Scale conviction with model confidence—uncertainty-weighted Kelly criterion

5.3 Limitations

Transaction costs are not modeled; frequent Bayesian rebalancing would incur higher trading costs. The backtest assumes perfect execution without slippage. Bitcoin's extreme returns dominate portfolio performance. The HMM assumes K=3 fixed states, which may not capture all market regimes.

6. Conclusion

This project demonstrated a comprehensive Bayesian framework for multi-asset dynamic portfolio allocation, integrating VAE, HMM, and BNN over an 11-year period (2015–2025). Key findings:

- All three Bayesian strategies outperform non-Bayesian counterparts: Sharpe improvements +0.21 to +3.43, return improvements +5.3% to +32.3%
- BayesianRiskParity (B) achieves the highest risk-adjusted performance: Sharpe 3.870, $100K → $4.16M with only 10.2% max drawdown
- Bayesian advantages persist out-of-sample: BayesianRiskParity achieves OOS Sharpe 4.726 with just −5.8% drawdown
- BayesianRiskParity (B) emerges as the best risk-adjusted OOS strategy (Sharpe 1.980, max DD −8.3%)

The core insight is that *knowing what you don't know* is as valuable as prediction accuracy in portfolio management. When BNN reports high uncertainty, reducing positions protects capital during regime transitions. This principle extends beyond finance: any decision system under non-stationary conditions benefits from principled uncertainty estimation.

7. References

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